

Introduction

Baroid Core Homework 21

Assessment Feedback

Congratulations, you passed.

Total score: 40 out of 40, 100%

Question Feedback

Question

1 of 27

Hole cleaning can be defined as the ability of the drilling fluid to transport drilled cuttings / cavings to surface or to keep them suspended during a static state

- ☒ True
☐ False

1 out of 1

Question

2 of 27

What are some of the good practices that can aid in hole cleaning?

- ☒ Increase circulation
☐ Avoid tripping
☐ High ROP
☒ Control fluid parameters
☒ Optimize mechanical separation equipment

3 out of 3

Question

3 of 27

An evidence of good hole cleaning is a buildup in solids inside the annular space, that leads to a stuck pipe situation

- ☐ True

- ☐ True
- ☒ False

1 out of 1

Question

4 of 27

Select some of the signs of inadequate hole cleaning

- ☒ Widely-fluctuating rotating torque
- ☒ Issues maintaining angle when sliding
- ☒ Stuck pipe
- ☒ Quick increase of ECD when pipe rotation is initiated
- ☒ Tight hole and /or pack-offs
- ☐ Decreased drag after connections
- ☐ All of the above

5 out of 5

Question

5 of 27

Inadequate hole cleaning can affect the drilling process, especially when building directional wells. The development of cutting beds can deviate the angle of the well while sliding

- ☒ True
- ☐ False

1 out of 1

Question

6 of 27

Indicate all the factors that can affect hole cleaning efficiency:

- ☒ Flow rate
- ☒ Drill pipe rotation
- ☒ Drilling rate (ROP)
- ☒ Drill pipe eccentricity
- ☐ Flow line temperature
- ☒ Drilling fluid density
- ☒ Angle of well
- ☒ Drilling fluid rheological properties
- ☐ Formation salinity
- ☒ Density, size & shape of cuttings

- ☐ Formation alkalinity
- ☐ Bottom hole pressures

8 out of 8

Question

7 of 27

As the angle of deviation of the wells increases, so does the difficulty in transporting the cuttings

- ☒ True
- ☐ False

1 out of 1

Question

8 of 27

In general, vertical wells are the hardest to clean due the effect of slip velocity

- ☐ True
- ☒ False

1 out of 1

Question

9 of 27

The formation of a cuttings bed is more pronounced when the angle of inclination of the well is greater than 30 degrees

- ☒ True
- ☐ False

1 out of 1

Question

10 of 27

For angles between 40° - 60°, a turbulent or transition flow and a low rheology fluid is the most effective combination for minimizing the formation of cuttings beds, removing cuttings beds, and cleaning high angle wells

- ☒ True
- ☐ False

1 out of 1

Question

11 of 27

Cuttings and cuttings beds will be removed more effectively under laminar flow

- ☐ True
- ☒ False

1 out of 1

Question

12 of 27

It has been demonstrated that in angles between 60° - 90°, high viscosity pills do not increase the hole cleaning efficiency, and can actually reduce the flow rates below the drill pipe, where bed cuttings accumulate

- ☒ True
- ☐ False

1 out of 1

Question

13 of 27

Flow rate can impact hole cleaning efficiency; however it needs to be controlled to avoid issues, such as? (select all that apply)

- ☒ Exceed capacity of the pumps
- ☒ Lead to stuck pipe
- ☒ Generate hole washout
- ☐ Maximize transportation of cuttings
- ☒ High ECD

1 out of 1

Question

14 of 27

An increase in fluid density will cause an increase in the buoyancy, thus improving the hole cleaning characteristics at any inclination.

1 out of 1

Question

15 of 27

The rheology of the drilling fluid should be carefully adjusted to maximize the transport capacity, minimize friction losses and minimize the impact on downhole pressures

- ☒ True
- ☐ False

1 out of 1

Question

16 of 27

The density, size and shape of the cuttings do not affect hole cleaning efficiency

- ☐ True
- ☒ False

1 out of 1

Question

17 of 27

Which of the following is harder to remove from the well?

- ☐ Lighter, medium, squared cutting
- ☒ Heavy, longer, rounder cutting
- ☐ Lighter, smaller, sharp cutting
- ☐ Heavy, longer, squared cutting

1 out of 1

Question

18 of 27

An increase in the rate of penetration will result in a greater concentration of cuttings in the annulus, which will improve the hole cleaning efficiency

- ☐ True
- ☒ False

1 out of 1

Question

19 of 27

Drill pipe rotation can have a pronounced effect upon cuttings transport by mechanically agitating a cuttings bed

- ☒ True
- ☐ False

1 out of 1

Question

20 of 27

When drilling a directional well, for hole cleaning purposes it's recommended to try to rotate the drill string as much as possible

- ☒ True
☐ False

1 out of 1

Question

21 of 27

The principal negative effect of drill pipe eccentricity in high-angle drilling is to sharply reduce fluid flow under the drill pipe where the annular gap is narrow

- ☒ True
☐ False

1 out of 1

Question

22 of 27

Experience has shown that a maximum average percentage of cuttings of 3% (for the entire well) should remain in the annulus, to avoid drilling issues

1 out of 1

Question

23 of 27

For each interval (diameter section) of the wellbore, a maximum of 10% may be handled without compromising the cleaning of the hole

1 out of 1

Question

24 of 27

Field studies indicate that the rotating the pipe while drilling improves the hole cleaning efficiency by approximately 5%.

- ☐ True

☒ False

1 out of 1

Question

25 of 27

For hole cleaning purposes, short trip while pumping are not recommended because they increase the time to drill the well and could disturb the cuttings beds in the wellbore

☐ True
☒ False

1 out of 1

Question

26 of 27

Sweeps are a primary tool to provide effective hole cleaning

☐ True
☒ False

1 out of 1

Question

27 of 27

It is recommended to pump only 2ppg over high density sweeps in deviated wells to avoid exceeding the fracture gradient of the wellbore?

☐ True
☒ False

1 out of 1